Supplementary material

This anonymous scientific supplement reports method contracts and aggregate evidence for the accompanying manuscript. It contains no author identity, employee-level record, fitted model, approval history, internal review, or development diary. Exact, unabridged versions of Tables S1–S4 and S6–S8 are supplied as CSV files in the `tables` directory.

Table S1. Feature-availability and governance matrix

`I` denotes included and `E` excluded. A question mark marks plausible pre-existence without a verified row-level timestamp. Full semantic descriptions, governance classes, justifications, and timestamp caveats appear in `tables/Table_S1.csv`.

Table S1a. Timing and risk classification

Feature	Timing	Risk
EmpNumber	Identifier	Identifier
Age	Pre-existing?	Sensitive
Gender	Pre-existing?	Sensitive
EducationBackground	Pre-existing?	Ordinary
MaritalStatus	Pre-existing?	Sensitive
EmpDepartment	Pre-existing?	Org. proxy
EmpJobRole	Pre-existing?	Org. proxy
BusinessTravelFrequency	Pre-existing?	Org. proxy
DistanceFromHome	Pre-existing?	Ordinary
EmpEducationLevel	Pre-existing?	Ordinary
EmpEnvironmentSatisfaction	Uncertain	Timing
EmpHourlyRate	Pre-existing?	Org. proxy
EmpJobInvolvement	Uncertain	Timing
EmpJobLevel	Pre-existing?	Org. proxy
EmpJobSatisfaction	Uncertain	Timing
NumCompaniesWorked	Pre-existing?	Ordinary
OverTime	Uncertain	Timing
EmpLastSalaryHikePercent	May postdate	Outcome-near
EmpRelationshipSatisfaction	Uncertain	Timing
TotalWorkExperienceInYears	Pre-existing?	Ordinary
TrainingTimesLastYear	Uncertain	Timing
EmpWorkLifeBalance	Uncertain	Timing
ExperienceYearsAtThisCompany	Pre-existing?	Ordinary
ExperienceYearsInCurrentRole	Pre-existing?	Org. proxy
YearsSinceLastPromotion	Pre-existing?	Org. proxy
YearsWithCurrManager	Pre-existing?	Org. proxy
Attrition	May postdate	temporal leakage
PerformanceRating	unavailable by design target	target direct leakage

Table S1b. Policy inclusion matrix

Feature	P0	P1	P2	P3	P4	P5
EmpNumber	E	E	E	E	E	E
Age	I	I	E	E	E	E
Gender	I	I	E	E	E	E
EducationBackground	I	I	I	I	I	I
MaritalStatus	I	I	E	E	E	E
EmpDepartment	I	I	I	E	E	E
EmpJobRole	I	I	I	I	I	E
BusinessTravelFrequency	I	I	I	I	I	E
DistanceFromHome	I	I	I	I	I	I
EmpEducationLevel	I	I	I	I	I	I
EmpEnvironmentSatisfaction	I	I	I	I	E	E
EmpHourlyRate	I	I	I	I	I	E
EmpJobInvolvement	I	I	I	I	E	E
EmpJobLevel	I	I	I	I	I	E
EmpJobSatisfaction	I	I	I	I	E	E
NumCompaniesWorked	I	I	I	I	I	I
OverTime	I	I	I	I	E	E
EmpLastSalaryHikePercent	I	E	E	E	E	E
EmpRelationshipSatisfaction	I	I	I	I	E	E
TotalWorkExperienceInYears	I	I	I	I	I	I
TrainingTimesLastYear	I	I	I	I	E	E
EmpWorkLifeBalance	I	I	I	I	E	E
ExperienceYearsAtThisCompany	I	I	I	I	I	I
ExperienceYearsInCurrentRole	I	I	I	I	I	E
YearsSinceLastPromotion	I	I	I	I	I	E
YearsWithCurrManager	I	I	I	I	I	E
Attrition	I	E	E	E	E	E
PerformanceRating	E	E	E	E	E	E

Table S2. Preprocessing and search contract

All preprocessing and candidate selection are confined to the current training partition. Exact estimator paths, fixed parameters, candidate grids, tolerances, and failure rules appear in `tables/Table_S2.csv`.

Type	Component	Candidates	Fit scope	Primary	Tie break	Outer test
preprocessing	Numeric preprocessing	0	Training partitions only	not applicable	not applicable	Evaluation only
preprocessing	Categorical preprocessing	0	Training partitions only	not applicable	not applicable	Evaluation only
trained model	Multinomial Logistic Regression	6	Training partitions only	macro f1	quadratic weighted kappa	Evaluation only
trained model	Random Forest	8	Training partitions only	macro f1	quadratic weighted kappa	Evaluation only
trained model	LightGBM	8	Training partitions only	macro f1	quadratic weighted kappa	Evaluation only
trained model	XGBoost	8	Training partitions only	macro f1	quadratic weighted kappa	Evaluation only
trained model	Proportional-Odds Ordinal Logistic Regression	6	Training partitions only	macro f1	quadratic weighted kappa	Evaluation only
trained model	Cumulative-Threshold XGBoost	8	Training partitions only	macro f1	quadratic weighted kappa	Evaluation only

Type	Component	Candidates	Fit scope	Primary	Tie break	Outer test
selection regime	macro f1 selection		Training partitions only	macro f1	quadratic weighted kappa; then lowest candidate index	Evaluation only
selection regime	qwk selection		Training partitions only	quadratic weighted kappa	macro f1; then lowest candidate index	Evaluation only

Table S3. Cross-validation overview

Exact seed schedules, fold identities, out-of-fold coverage, calibration isolation, selection-score sources, and reuse/refit boundaries appear in `tables/Table_S3.csv`.

Analysis	Population	n	Outer design	Inner design	Outer test
Canonical benchmark	INX/P3	1	10-fold stratified	5-fold stratified	Evaluation only
Repeated-CV sensitivity	INX/P3	5	5-fold stratified	5-fold stratified	Evaluation only
Fixed-policy sensitivity	INX/P0-P5	1	10-fold stratified	5-fold stratified	Evaluation only
Retuned-policy sensitivity	INX/P0-P5	1	10-fold stratified	5-fold stratified	Evaluation only
Selection-objective sensitivity	INX/P3	1	10-fold stratified	5-fold stratified	Evaluation only
Sigmoid calibration	INX/P3/XGBoost	1	10-fold stratified	Persisted folds	Evaluation only
HR target/CV sensitivity	HRDataset v14/conservative seven features	5	5-fold stratified	5-fold stratified	Evaluation only
HR target-alias sensitivity	HRDataset v14/311 historical and matched 309	1	5-fold stratified	5-fold stratified	Evaluation only

Table S4. Per-class performance by selection objective

Selection	System	Rating	Precision	Recall	F1	Support
macro f1	cumulative threshold xgboost	2	0.7974	0.9330	0.8599	194
macro f1	cumulative threshold xgboost	3	0.8700	0.6739	0.7595	874
macro f1	cumulative threshold xgboost	4	0.1858	0.4167	0.2570	132
macro f1	lightgbm	2	0.8233	0.9124	0.8655	194
macro f1	lightgbm	3	0.8497	0.8924	0.8705	874
macro f1	lightgbm	4	0.1194	0.0606	0.0804	132
macro f1	logistic regression	2	0.5485	0.6701	0.6032	194
macro f1	logistic regression	3	0.8010	0.7277	0.7626	874
macro f1	logistic regression	4	0.1361	0.1742	0.1528	132

Selection	System	Rating	Precision	Recall	F1	Support
macro f1	proportional odds logistic	2	0.5126	0.7320	0.6030	194
macro f1	proportional odds logistic	3	0.7920	0.5183	0.6266	874
macro f1	proportional odds logistic	4	0.1538	0.4091	0.2236	132
macro f1	random forest	2	0.8364	0.9227	0.8775	194
macro f1	random forest	3	0.8517	0.9531	0.8996	874
macro f1	random forest	4	0.0000	0.0000	0.0000	132
macro f1	xgboost	2	0.8211	0.9227	0.8689	194
macro f1	xgboost	3	0.8542	0.8112	0.8322	874
macro f1	xgboost	4	0.1513	0.1742	0.1620	132
qwk	cumulative threshold xgboost	2	0.8447	0.8969	0.8700	194
qwk	cumulative threshold xgboost	3	0.8491	0.9657	0.9036	874
qwk	cumulative threshold xgboost	4	0.0000	0.0000	0.0000	132
qwk	lightgbm	2	0.8293	0.8763	0.8521	194
qwk	lightgbm	3	0.8460	0.9428	0.8918	874
qwk	lightgbm	4	0.1905	0.0303	0.0523	132
qwk	logistic regression	2	0.5375	0.6649	0.5945	194
qwk	logistic regression	3	0.7985	0.7208	0.7577	874
qwk	logistic regression	4	0.1345	0.1742	0.1518	132
qwk	proportional odds logistic	2	0.4836	0.8351	0.6125	194
qwk	proportional odds logistic	3	0.8121	0.4005	0.5364	874
qwk	proportional odds logistic	4	0.1567	0.5152	0.2403	132
qwk	random forest	2	0.8249	0.9227	0.8710	194
qwk	random forest	3	0.8520	0.9554	0.9008	874
qwk	random forest	4	0.0000	0.0000	0.0000	132
qwk	xgboost	2	0.8524	0.9227	0.8861	194
qwk	xgboost	3	0.8535	0.9600	0.9036	874
qwk	xgboost	4	0.1429	0.0076	0.0144	132

Hard predictions use the declared argmax rule with no class-specific threshold optimization. The exact table is `tables/Table_S4.csv`.

Table S5. Complete audit gates

Stage	Input	Audit	Failure	Permissible claim
Information	Features and intended context	Availability/timing contract	Timing unverified	Sensitivity only
Selection	Outer-training partitions	Nested candidate selection	Test-informed selection	Evaluation invalid
Prediction	Untouched outer test	OOF probabilities; ordinal/class metrics	Extreme-class collapse	Bound aggregate claim

Stage	Input	Audit	Failure	Permissible claim
Calibration	Cross-fitted training probabilities	Predeclared sigmoid fit	Outer-test fit/selection	Calibration invalid
Explanation	Prediction-producing fold model	Identity, additivity, stability, deletion	Model mismatch	No explanation claim
Subgroup/proxy	Held-out outputs/categories	Support, interval, distinct proxy questions	Unsupported denominator	Descriptive/not estimated
Replication	Separately fitted HR systems	Target, folds, calibration, support	Construct/transport conflation	Partial replication
Evidence	Aggregate source rows	Selector, value, hash, qualifier	Unresolved/stale number	No numerical claim

Table S6. Subgroup gap uncertainty and support

Intervals are 95% simultaneous exploratory intervals based on the studentized maximum absolute bootstrap deviation over all estimable prespecified P3 attribute, support-threshold, and metric gap cells. Eligibility is fixed before resampling; fitted-model variability is excluded. These diagnostics do not certify fairness.

Age

Metric	Gap	Minimum group	Maximum group	Simultaneous low	Simultaneous high
macro f1	0.0301	30-39	40-49	0.0000	0.1657
quadratic weighted kappa	0.0653	30-39	50-59	0.0000	0.3335
ordinal mae	0.0770	50-59	30-39	0.0000	0.2411

Gender

Metric	Gap	Minimum group	Maximum group	Simultaneous low	Simultaneous high
macro f1	0.0363	Male	Female	0.0000	0.1515
quadratic weighted kappa	0.0527	Male	Female	0.0000	0.2420
ordinal mae	0.0508	Female	Male	0.0000	0.1719

Marital Status

Metric	Gap	Minimum group	Maximum group	Simultaneous low	Simultaneous high
macro f1	0.0101	Married	Divorced	0.0000	0.1135
quadratic weighted kappa	0.0381	Single	Married	0.0000	0.2431
ordinal mae	0.0291	Divorced	Single	0.0000	0.1485

Business Travel

Metric	Gap	Minimum group	Maximum group	Simultaneous low	Simultaneous high
macro f1	0.0528	Travel Frequently	Travel Rarely	0.0000	0.1987
quadratic weighted kappa	0.0711	Travel Frequently	Travel Rarely	0.0000	0.3472
ordinal mae	0.0613	Travel Rarely	Travel Frequently	0.0000	0.2195

Department

Metric	Gap	Minimum group	Maximum group	Simultaneous low	Simultaneous high
macro f1	0.2179	Development	Finance	0.0000	0.5281
quadratic weighted kappa	0.4388	Development	Research & Development	0.1275	0.7501
ordinal mae	0.1193	Finance	Development	0.0000	0.3687

Education Background

Metric	Gap	Minimum group	Maximum group	Simultaneous low	Simultaneous high
macro f1	0.0413	Other	Medical	0.0000	0.3766
quadratic weighted kappa	0.1190	Technical Degree	Marketing	0.0000	0.5064
ordinal mae	0.0891	Medical	Technical Degree	0.0000	0.3211

The exact support counts, pointwise intervals, resample counts, source identities, and claim boundaries appear in `tables/Table_S6.csv`.

Table S7. Selection-objective confusion matrices

Selection	System	True	Predicted	Count
macro f1	cumulative threshold xgboost	2	2	181
macro f1	cumulative threshold xgboost	2	3	13
macro f1	cumulative threshold xgboost	2	4	0
macro f1	cumulative threshold xgboost	3	2	44
macro f1	cumulative threshold xgboost	3	3	589
macro f1	cumulative threshold xgboost	3	4	241
macro f1	cumulative threshold xgboost	4	2	2
macro f1	cumulative threshold xgboost	4	3	75
macro f1	cumulative threshold xgboost	4	4	55
macro f1	lightgbm	2	2	177
macro f1	lightgbm	2	3	16
macro f1	lightgbm	2	4	1
macro f1	lightgbm	3	2	36
macro f1	lightgbm	3	3	780
macro f1	lightgbm	3	4	58
macro f1	lightgbm	4	2	2
macro f1	lightgbm	4	3	122
macro f1	lightgbm	4	4	8
macro f1	logistic regression	2	2	130
macro f1	logistic regression	2	3	59

Selection	System	True	Predicted	Count
macro f1	logistic regression	2	4	5
macro f1	logistic regression	3	2	97
macro f1	logistic regression	3	3	636
macro f1	logistic regression	3	4	141
macro f1	logistic regression	4	2	10
macro f1	logistic regression	4	3	99
macro f1	logistic regression	4	4	23
macro f1	proportional odds logistic	2	2	142
macro f1	proportional odds logistic	2	3	52
macro f1	proportional odds logistic	2	4	0
macro f1	proportional odds logistic	3	2	124
macro f1	proportional odds logistic	3	3	453
macro f1	proportional odds logistic	3	4	297
macro f1	proportional odds logistic	4	2	11
macro f1	proportional odds logistic	4	3	67
macro f1	proportional odds logistic	4	4	54
macro f1	random forest	2	2	179
macro f1	random forest	2	3	15
macro f1	random forest	2	4	0
macro f1	random forest	3	2	33
macro f1	random forest	3	3	833
macro f1	random forest	3	4	8
macro f1	random forest	4	2	2
macro f1	random forest	4	3	130
macro f1	random forest	4	4	0
macro f1	xgboost	2	2	179
macro f1	xgboost	2	3	14
macro f1	xgboost	2	4	1
macro f1	xgboost	3	2	37
macro f1	xgboost	3	3	709
macro f1	xgboost	3	4	128
macro f1	xgboost	4	2	2
macro f1	xgboost	4	3	107
macro f1	xgboost	4	4	23
qwk	cumulative threshold xgboost	2	2	174
qwk	cumulative threshold xgboost	2	3	20
qwk	cumulative threshold xgboost	2	4	0
qwk	cumulative threshold xgboost	3	2	30
qwk	cumulative threshold xgboost	3	3	844
qwk	cumulative threshold xgboost	3	4	0
qwk	cumulative threshold xgboost	4	2	2
qwk	cumulative threshold xgboost	4	3	130
qwk	cumulative threshold xgboost	4	4	0
qwk	lightgbm	2	2	170
qwk	lightgbm	2	3	24
qwk	lightgbm	2	4	0
qwk	lightgbm	3	2	33
qwk	lightgbm	3	3	824
qwk	lightgbm	3	4	17
qwk	lightgbm	4	2	2
qwk	lightgbm	4	3	126
qwk	lightgbm	4	4	4
qwk	logistic regression	2	2	129
qwk	logistic regression	2	3	60
qwk	logistic regression	2	4	5
qwk	logistic regression	3	2	101
qwk	logistic regression	3	3	630
qwk	logistic regression	3	4	143
qwk	logistic regression	4	2	10
qwk	logistic regression	4	3	99
qwk	logistic regression	4	4	23
qwk	proportional odds logistic	2	2	162
qwk	proportional odds logistic	2	3	32
qwk	proportional odds logistic	2	4	0
qwk	proportional odds logistic	3	2	158
qwk	proportional odds logistic	3	3	350
qwk	proportional odds logistic	3	4	366
qwk	proportional odds logistic	4	2	15
qwk	proportional odds logistic	4	3	49
qwk	proportional odds logistic	4	4	68
qwk	random forest	2	2	179
qwk	random forest	2	3	15

Selection	System	True	Predicted	Count
qwk	random forest	2	4	0
qwk	random forest	3	2	36
qwk	random forest	3	3	835
qwk	random forest	3	4	3
qwk	random forest	4	2	2
qwk	random forest	4	3	130
qwk	random forest	4	4	0
qwk	xgboost	2	2	179
qwk	xgboost	2	3	15
qwk	xgboost	2	4	0
qwk	xgboost	3	2	29
qwk	xgboost	3	3	839
qwk	xgboost	3	4	6
qwk	xgboost	4	2	2
qwk	xgboost	4	3	129
qwk	xgboost	4	4	1

The exact table, including the dataset key, is `tables/Table_S7.csv`.

Table S8. Bounded literature comparison

Li et al. (2021)

Problem and evidence: Employee-performance classification; Three linked internal HR tables from a publishing organization. **Method and XAI:** Logistic regression; decision tree; naive Bayes; Permutation importance reported. **Boundary relative to this study:** Organization-specific comparison; present work joins ordinal failure analysis to an explicit evidence contract. Official proceedings PDF; no claim of an exhaustive absence audit.

Adeniyi et al. (2022)

Problem and evidence: Employee-performance classification; Public HR table with 311 rows and 36 attributes. **Method and XAI:** PCA; ANN; RF; DT; reported 80:20 split; not_reported_in_accessible_source. **Boundary relative to this study:** Present work separately audits target aliases, target mapping and training-only evaluation. Publisher full text. Existing title retained over conflicting Crossref deposit.

Putri and Sitohang (2026)

Problem and evidence: Employee-performance classification; INX employee data. **Method and XAI:** Random Forest; multiple train/test proportions; Feature importance. **Boundary relative to this study:** Present work separates nested candidate selection from final evaluation and audits metric-specific extreme-class failures. Publisher PDF methods/results; non-reporting is not proof of absence.

Löfström et al. (2024)

Problem and evidence: Reliable local explanations with uncertainty; 25 benchmark datasets. **Method and XAI:** Calibrated Explanations based on Venn-Abers; Conditional feature explanations and counterfactuals. **Boundary relative to this study:** A new explanation method; present study audits existing exact-model SHAP and a declared probability-calibration path. Publisher abstract and metadata; do not infer unseen evaluation controls.

Sepúlveda et al. (2025)

Problem and evidence: Stability of local interpretability; Synthetic and real anomaly-detection data. **Method and XAI:** Top-rank-weighted stability measure; SHAP-based feature-ranking consistency under small changes. **Boundary relative to this study:** Local perturbation stability differs from fold/seed/training-resample comparisons and model-level deletion in the present audit. Publisher abstract and section excerpts.

Manafi Varkiani et al. (2025)

Problem and evidence: Employee attrition and contextual explanation; Italian financial-company employee records. **Method and XAI:** Machine-learning case study; Random Forest explanation; SHAP contribution magnitude and direction. **Boundary relative to this study:** Strong HR-XAI precedent; different outcome and organization, without evidence of the present shared audit contract in accessible material. Publisher abstract and available methods excerpts.

Rass and Dallinger (2026)

Problem and evidence: Detecting unwanted patterns in training data; Real-life and synthetic datasets. **Method and XAI:** Prespecified rules; fuzzy reasoning; regression significance tests; Rule-based explainability. **Boundary relative to this study:** Proposes data-testing machinery; present study evaluates how multiple system-audit views constrain claims on fixed HR evidence. Publisher abstract and metadata; final publication year 2026 despite DOI containing 2025.

Abusitta et al. (2024)

Problem and evidence: Taxonomy and limitations of XAI; Literature survey. **Method and XAI:** Review and comparative taxonomy; Central topic. **Boundary relative to this study:** General XAI guidance; present paper must show a concrete operational audit rather than treat a component list as novelty. Publisher abstract and section excerpts; not a benchmark competitor.

Kapoor et al. (2024), REFORMS

Problem and evidence: Reporting and reproducibility of ML-based science; Consensus and methodological literature. **Method and XAI:** Checklist and guidelines; Reporting context. **Boundary relative to this study:** Important prior protocol. Present contribution should be the executable linkage plus empirical interactions among audit findings, not the invention of transparent evaluation. Author-maintained project and registered Science Advances metadata.

Bar-Gil et al. (2024)

Problem and evidence: Embedding AI ethics in HR analytics; Two organizational HR case studies. **Method and XAI:** Comparative case-study analysis; Organizational governance context. **Boundary relative to this study:** Shows why technical diagnostics cannot settle organizational governance or employee consequences. Publisher abstract and full-text excerpts.

Lones (2024)

Problem and evidence: Avoiding common ML practice errors; Tutorial and examples. **Method and XAI:** Practical methodological guidance; Interpretation guidance. **Boundary relative to this study:** Broad practical precedent; present study instantiates connected checks in an ordinal HR setting. Publisher tutorial identity and abstract; individual empirical comparisons are not inferred.

Present study

Problem and evidence: Auditing ordinal employee-performance prediction; INX; partial HRDataset_v14 protocol replication. **Method and XAI:** Nested benchmark; objective/policy sensitivities; exact evidence identities; Exact-model grouped SHAP; stability; model-level deletion. **Boundary**

relative to this study: Integration links conflicting findings to the evaluated system; no new predictor or deployment claim. Frozen claim boundary; not an additional experimental result.

The full comparison dimensions and official-source links appear in `tables/Table_S8.csv`.

Numerical evidence ledger

The machine-readable ledger and its twenty anonymous aggregate source tables are supplied in the `evidence` directory. `SOURCE_INDEX.csv` records every anonymous source hash. The ledger preserves exact values, selectors, rounding, required qualifiers, and prohibited interpretations for the 99 active numerical claims.